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ORIGINAL ARTICLE

Additive friction stir deposition induced stress ripples in aluminum alloy

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Abstract
In additive manufacturing, modification of the viscosity of softened or molten metal plays a significant role in the development of mechanical properties distribution. Similar to soft polymers, molten or softened metal usually exhibits vastly different viscoelastic properties. During scratching tests, viscoelastic polymers undergo spatiotemporal stress development and relaxation, causing stress ripples to form on their surfaces. It is observed that friction stir-based solid-state manufacturing can yield similar stress ripples on softened aluminum alloys due to the linear motion of the deposition tool. Softened aluminum alloys exhibit viscoelastic and viscoplastic responses similar to polymeric material during the solid-state deposition process. In the final product, residual stress ripples appear on the surface of the metal alloy due to periodic residual stress relaxation, leading to oscillatory behavior from the bulk mechanical properties of the material. The orientation of the in-built stress pattern formed in the bulk of the additively manufactured metal alloy underneath the visually evident onion ring-like surface pattern formation is diametrically in the opposite direction. The experimentally observed results are consistent with analytical calculation and numerical simulations in terms of behaviors.

Keywords Friction stir · Viscoelasticity · Scratch ripple · Ultrasound elastography · Softened aluminum

1 Introduction

Friction stir-based solid-state manufacturing techniques are rapidly evolving for the development of novel metal alloys and hybrid materials [1]. Friction stir processing results in rotational friction-induced heating that can lead to three-quarters of the melting point of metal alloys. During mechanical processing, softened materials are subjected to extreme plastic deformation and built residual stress, which may lead to significant changes in their mechanical properties. These plastic-deformation-based modifications enable the advanced processes of aluminum and magnesium alloys due to relatively high effective viscosity that is not usually achievable by fusion-based methods [2]. The high-speed rotation of the tool and induction of the feeding material generated frictional heat that softened the feeding material and facilitated plastic deformation [3]. Due to the normal downward force from the feeding material and the tool, robust plastic deformation reshaped the softened material into a thinner layered structure on a rigid substrate. A translational motion associated with the rotation of the tool deposits material along the track and deposits the metal alloy additively as a part of the solid-state manufacturing process. As in the case of viscoelastic polymer materials, the enhanced viscosity of the softened metallic alloy results in a complex evolution of the bulk properties and characteristics of the final built material [4–9]. . During friction stir-based manufacturing, this complexity is in terms of the mechanical characteristics and microstructure evolution associated with plastic deformation and material processing. This current work analyzes the formation and relaxation of the stress in the system due to the temporal change in the viscoelastic properties of the built material during the

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solid-state manufacturing process using the model developed in the polymeric material system.

Friction stir-based processes combine the rotational and linear translation motion of the tool. The tool rotation-induced shear strain effects were commonly theoretically discussed and numerically predicted in the existing literature, as discussed in the preceding discussions [10–12]. On the contrary, the linear translational motion-induced effects are barely discussed. The strain rate of linear translational motion is much slower compared to the rotation motion. In the case of viscoelastic material, the damping relaxation rate is comparable to the strain rate from the tool's linear motion during processing. In such cases, the softened materials are significantly sensitive to the scratching behaviors following the inverse-proportional relation between the factor E_{eff}/σ_{yield}, and the scratching resistance [13, 14]. During the solid-state friction stir-based processes, the softened solid-state aluminum alloys have a significantly decreased yield strength, σ_{yield}, with respect to an insignificantly reduced effective elastic modulus, E_{eff} [15, 16]. It leads to a high sensitivity to the scratching characteristics of the softened aluminum alloys.

Such scratching behaviors on the softened aluminum result in residual stresses along the main depth direction on the surface and subsurface region of the final products. Due to the polishing-induced residual stress relaxing, conventional mechanical tests and microscopic observations meet challenges in performing the sample preparation processing [17]. Hence, the alternative solution was selected to use non-destructive ultrasound elastography to map out the distributions of the stress patterns in terms of frequency-dependent dynamic (effective) bulk modulus K_d [18–20]. In the fundamental relation of bulk modulus, the externally applied stress on the medium can be balanced by the varying atomic force with the change of atomic distance [5, 21], which makes dynamic bulk modulus elastography applicable to present the distribution of residual stress by its variations. In this study, the viscoelastic damping effects induced residual stress patterns were explained, simulated, and observed by viscoelastic scratching theory, finite element analysis, and ultrasound elastographic mapping.

2 Materials and methods

Finite element analysis (FEA)-based software, COMSOL Multiphysics, was selected to perform numerical studies for observation on the scratching behaviors only. In the 3D models, a multi-physic linked heat transfer module and a solid mechanics module (Johnson-Cook model) were used. Based on the existing literature [22, 23], the heating source described friction stir-induced heat generation from the tool's shoulder. The heat source was under moving mesh and rotation frame conditions. The interface was coupled as the pair condition. For time-dependent investigations, the model was re-meshed between each time interval based on the results from the previous time point. On the substrates, the four side walls were free. The upper and bottom surfaces were under fixed boundary conditions for observing residual stress patterns. Natural convection cooling with room temperature air was assigned to the upper surface and for the sidewalls of the substrate. The linear motion was performed by pre-described displacement on the heat source under constant moving speed. The viscoelastic properties of 6xxx aluminum can be found in the literature [24–26].

The reference aluminum alloy 6061-T6 plates and feed rods were purchased commercially for experimental studies. The additive friction stir deposition was performed on a MELD machine. The program to create the desired deposition path was written and run on the machine's CNC controller. The feed rod is loaded into the center bore of the deposition tool. The tool is rotated in the machine, and the tool face is slowly plunged into the substrate material. The heat generated by friction (Fig. 1A) from the shear stirring action of the tool softens the feed rod (Fig. 1B), which is forced out of the tool face and into the substrate by the machine actuator, fully plasticization the material and extruding the deposition layer. At this point, the linear motion begins along the programmed path to perform the deposition (Fig. 1C). In fabricating all examined samples, the tool rotation speed was 200 rpm, feed rod velocity was 152 mm/min, and tool height was about 1 mm.

The schematic experimental setup is illustrated in Fig. 1D. The ultrasound-based elastography (EBME) tests were performed in ambient air conditions at room temperature. A non-contact transducer with a 2.1 MHz nominal frequency from the Ultran Group company was used to experiment. The active diameter of the transducer is 16 mm. A UR5e collaborative robotic arm from Universal Robots performed a raster scan with a predefined script from MATLAB. A JSR Ultrasonic DPR 500 pulse/receiver operated the pulse source and time trigger. A Tektronix MDO 34 oscilloscope was used to collect data with a 1 GHz sampling rate. The scanned area has a spatial resolution of 0.5 mm.

3 Results and discussion

Under a stress σ, the strain behavior ϵ of ideal elastic materials corresponds to the spring model that follows Hooke's law, $\sigma = E\epsilon$, where E is the elasticity tensor. In soft or softened materials, additional damping behaviors present on the strain/ stress response following $\sigma = \eta\frac{d\epsilon}{dt}$, where η refers to the effective viscosity of the materials. Such strain /stress behaviors were called viscoelasticity response, which following the relation [27, 28]:

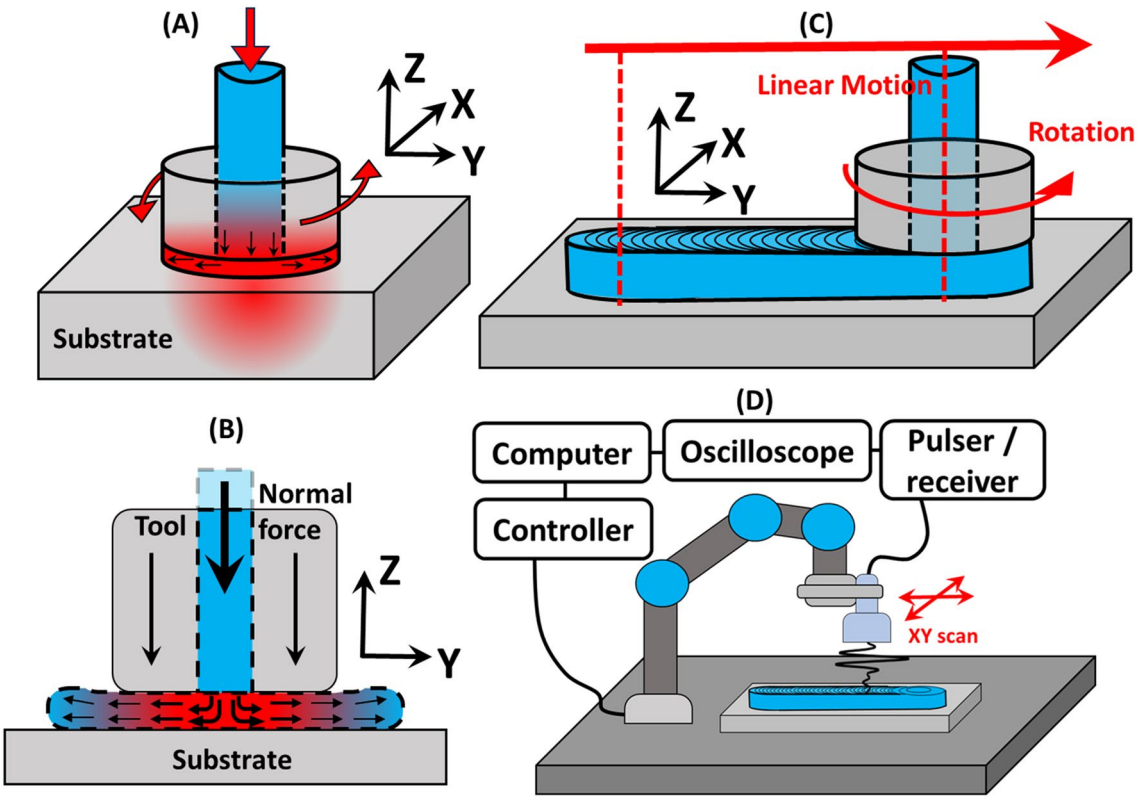

Fig. 1 Illustration of the additive friction stir deposition processing. **A** A high rotation speed induces frictional heating effects on the friction stir tool and feeding material. **B** Actuated force (normal force on the feed material) and the normal force from the tool apply plastic deformation on the thermal softened metallic material heated by frictional heating. **C** The linear translation motion associated with rotational motion enables the plastic deformation to be spatially continuous during the deposition of an additional metal layer. **D** Experimental configuration of ultrasound elastography

$$\sigma(t) = E_{eff}\epsilon(t) + \int_0^t R(t-t\prime)\dot{\epsilon}\left(t'\right)dt\prime \tag{1}$$

where t is time, $R(t)$ is the relaxation function, and E_{eff} is the frequency-dependent elastic, also known as the complex modulus or dynamic modulus. In a linear model, the damping effect can be treated in conjunction or separately from the Maxwell viscoelasticity model, $\sigma + \frac{\eta}{E}\dot{\sigma} = \eta\dot{\epsilon}$, or Kelvin-Voigt viscoelasticity model, $\sigma = E\epsilon + \eta\dot{\epsilon}$. Due to the finite and noticeable relaxation time from the viscoelasticity response, periodical responses were expected to be caused by stresses under a long enough displacement.

Viscoelastic properties are important parameters for understanding the high strain rate mechanical and acoustic responses in soft polymer fields [29, 30]. However, the viscoelastic behaviors in softened metallic materials are receiving increasing interest in friction stir (FS)-based aluminum alloy processing. Friction stir-based processing operates and modifies the softened aluminum alloys to around 70–85% of its melting point. In such operating temperature ranges, the aluminum alloys are fluid-like and sticky, and the effective viscosity exhibits increased sensitivity to temperature and strain rate dependence. The simplified model describes the softened aluminum alloys following the proportion of power law dynamic viscosity [24–26]:

$$\eta\left(\dot{\epsilon}_{shear}, T\right) \propto e^{T_0 - T}\dot{\epsilon}_{shear}^{n-1} \tag{2}$$

where T_0 is melting temperature, and n is a scaling factor that can be smaller or greater than 1, indicating the inverse-proportional or proportional dependence between the shear strain rate and effective viscosity. In aluminum alloys, both shear-thinning and shear-sticking factors are considered in the literature [31–33]. In the case of friction stir processing, the inverse-proportionality relation was assumed to be an appropriate model. To further involve the viscoelasticity, the effective viscosity of aluminum alloys, especially AA6xxx, can be fitted into or modeled by the Carreau model [34], Perzyna model [35], or Bingham model [31]. Such studies demonstrated the possibility of predicting or analyzing the material flow fields for the friction stir-based processes for parameter selection and optimization. In AFSD, the heating temperature T is mainly contributed to by the tool rotation-induced friction. Furthermore, the shear strain rate

$\dot{\epsilon}_{shear}$ became the major variable resulting in the different effective viscosity-induced viscoelastic damping effects. In the shear strain rate, including rotational motion shear and linear motion shear, the rotational motion variation was limited due to the need for consistent heat generation. It is also less related to linear "scratching" behavior. Hence, the AFSD linear speed will be considered the selected variable in the following work to analyze and observe the change in the effective viscosity-induced viscoelastic damping effect.

In the present study, additive friction stir deposition (AFSD) demonstrates the linear translation motion-induced "scratching" behavior on softened aluminum 6061. Assuming the operating tool rotation speed to be downward, the normal stress is consistent in all the theoretical discussions in this study. The scratching stress σ_{sc}, or the tangential stress, is inversely proportional to the effective viscosity $\eta\left(\dot{\epsilon}_{shear}, T\right)$, which is proportional to the temperature during the processing. It is worth noting that the effective viscosity plays a crucial role in distinguishing between scratching behaviors and sliding behaviors. Therefore, by analyzing the scratching stress σ_{sc} on friction-stirred products, one can infer and compare the effective viscosity on a relative scale. Such retrieval of in situ behavior for ex situ analysis is unusual in industrial inspections and strongly motivated the investigation of scratching behaviors in friction-stir-based processes.

In viscoelastic materials, the scratching behaviors are initially due to the accumulation of deformation or stress along the linear motion. In the presence of damping effects, the strain rate-dependent deformation period of a damper can lead to an appropriate relaxation time. The relaxation of the accumulated deformation that occurs can be released as a stress/strain and residual stress/strain. The relaxation time t_R is finite and inversely proportional to the scratching speed. The linear motion speed in friction stir-based processes, $t_R \propto V_{lin}{}^{-1}$ (Fig. 2A) [36]. In other words, in a long enough friction stir deposition track, $t_{deposition} \gg t_R$, the periodic residual scratching stress or residual scratching strain can be present. It is because the stress/strain frequency is proportional to the linear motion speed, i.e., $f_{ripple} \propto V_{lin}$ (Fig. 2A), commonly termed ripples in viscoelastic polymer studies.

As previously mentioned, the initiation of the scratching ripples is based on the accumulating deformation or stress. Hence, the curvature direction of the scratching ripples is presented with the front curvature of the friction stir tool. The detailed expression of the ripple configuration is not currently derived in the literature due to its strong material dependence [37, 38]. Instead of an analytical solution, the approximation of the ripple radius is commonly performed by using Hertzian theory described stress patterns [39, 40]. As Fig. 2B illustrates, the radius of the ripple curvature $R_{Pattern}$ follows the proportion [41]:

$$\frac{1}{R_{Pattern}} \propto \frac{1}{R_{tool}} - \frac{F_n}{E_{eff} a^3} \tag{3}$$

where F_n is the downward normal force, and a refers to the contacting area between the scratching tool and the scratched material, which is highly dependent on the ratio between the feeding speed and linear motion speed in AFSD. At the extreme condition, the width of the deposited material w_{sample} is comparable to the tool diameter $2R_{tool}$, the maximum contacting area is approaching the area of the tool, $\pi R_{tool}{}^2$. With the fast linear motion speed, the contacting area a is smaller than $\pi R_{tool}{}^2$, but it is inversely proportional to the linear motion speed with a consistent material feeding rate. Hence, Fig. 2B presents the proportional behavior between linear motion speed and the radius of the ripple curvature $R_{Pattern}$. As the inserted subfigures illustrated, the radius of the ripple curvature can be smaller or greater than the radius of the tool with significant slow or fast linear translation speed, respectively.

On the longitudinal direction, the linear motion speed further tuned the stress profile (ripple profile) and the corresponding periodicity, where the amplitude of stress roughly follows a linear proportion between linear strain rate $\dot{\epsilon}$ and $\ln(\sigma)$ [42]. Due to the damping effect in the viscoelastic model, the strong strain rate dependence needs to be considered in the ripple profile estimations. Within one period time of ripple formation, the loading and unloading behaviors of viscoelastic materials follow the G'sell-Jonas equation [43]. By replacing the linear strain ϵ with the product of linear strain rate $\dot{\epsilon}$ and scratching time t, the strain rate-dependent ripple profiles (Fig. 2C) can be estimated with consistent operating temperature following the expression [13]:

$$\sigma(t) \propto (1 - e^{-\dot{\epsilon}t})\dot{\epsilon}^m e^{h(\dot{\epsilon}t)^2} \tag{4}$$

where original temperature gradient induced deformation $e^{(\alpha_T/T)}$ was not included in the estimation in Fig. 2C [13]; m and h are the strain rate sensitivity factor and strain hardening coefficient which can be obtained from the literature for aluminum alloys [22, 44]. As the strain rate increases, AFSD tool linear motion increases in the work; the length of a single stress ripple is inversely proportional to the tool's linear speed (Fig. 2C).

In a normalized time period, the strain rate-dependent ripple profiles are summarized and compared in Fig. 2C. On the profiles, the curves suggested the tool scratched from left to right. From the cooler color profile to the warmer color profile, the increased strain rate and linear motion speed lead to a phase delay and broadening effects on the ripple profiles, as the inserted subfigure illustrated. Regarding the absolute amplitude, the peak stress increases along with increasing linear motion speed. In addition, the advancing edge of the ripple profiles shows rapid slopes compared to the profiles

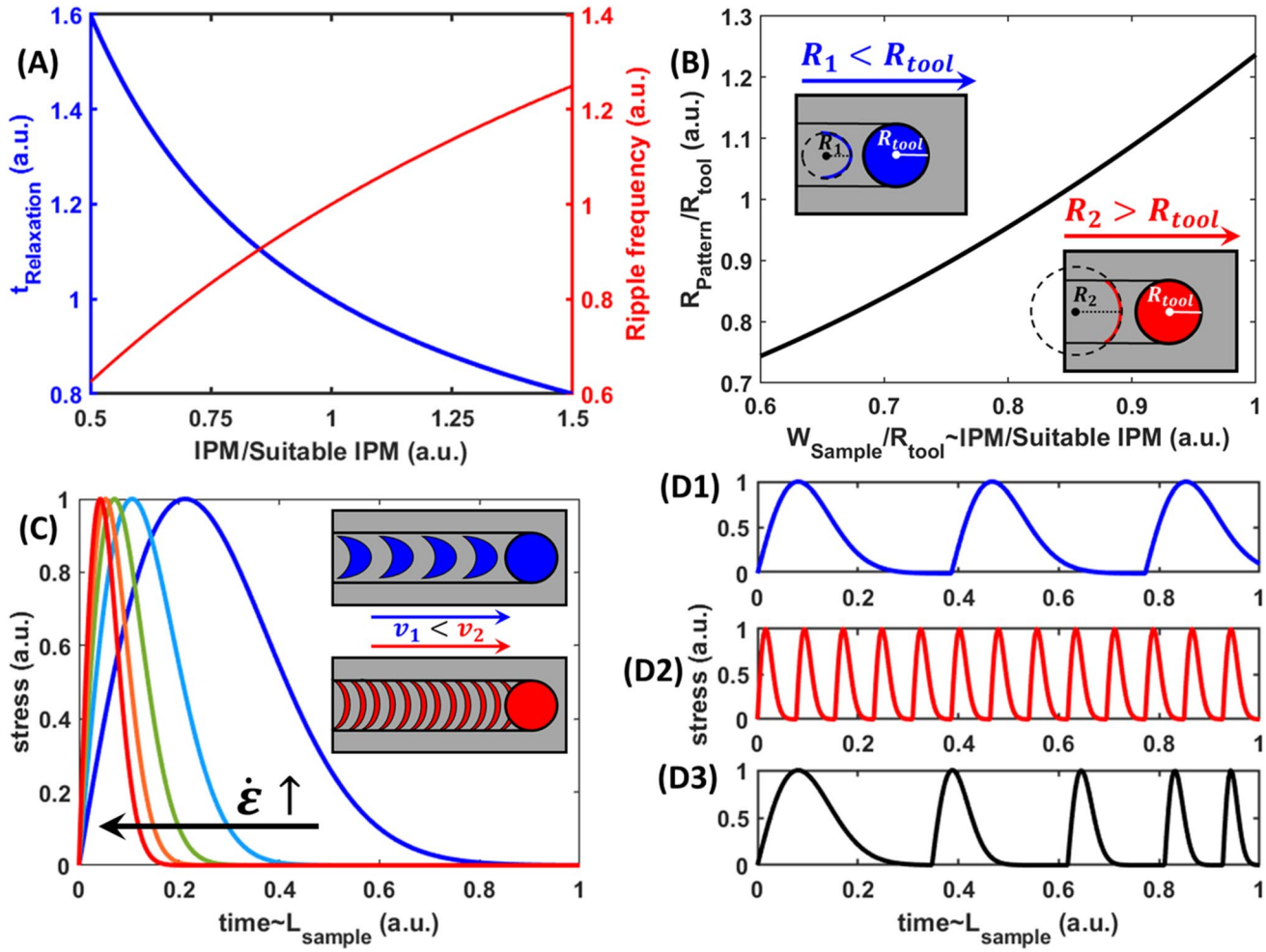

Fig. 2 **A** Linear motion speed ("scratching velocity") versus the viscoelastic relaxation time and ripple frequency of the loaded damping-spring model. IPM refers to an inch per minute in AFSD processing. The horizontal axis is normalized to have over-speeding (IPM/Suitable IPM > 1) linear motion and too-slow linear motion (IPM/Suitable IPM < 1). **B** Linear motion speed ("scratching velocity") dependent viscoelastic ripple radius shows a positive proportion with a constant tool radius. **C** Linear motion speed (linear strain rate) dependent width (length on the linear motion direction) of viscoelastic ripple radius shows a negative proportion under normalized amplitudes. **D1** and **D2** Linear motion speed relates frequency and width estimations stress ripples in long enough samples with slow and fast linear speeds, respectively. **D3** Gradient increased linear motion speed induces a decrease in the width of the stress ripples and an increase in the frequency of presence of the ripples

of the trailing edges. The slopes of the ripple profiles also increase proportionally relative to the rise of the linear translation speed.

As mentioned, the ripple frequency is proportional to the linear motion speed, which refers to the number of ripples increasing along with increasing linear speed during a certain long enough period, as Fig. 2D1 and D2 illustrates [45]. The profiles refer to the tool scratched on the softened aluminum alloys from the left side to the right side. The total time length between the different strain rate cases remained identical in the comparison. In addition, an additional ripple profile is calculated in Fig. 2D3 using a gradient increasing linear translation velocity. As a result, the ripple profile showed a softening behavior on both stress distribution and the frequency of the ripple profiles, which is usually observed in polymers [46].

After the overview of the theoretical behaviors of the longitudinal viscoelastic behaviors on the softened aluminum alloys in friction stir-based processes, practical observations were planned to be obtained via friction stir-based processed products. From various friction stir-based metal processing techniques, additive friction stir deposited aluminum alloys provide more apparent observation due to their large "stir zone" sizes. Figures 3 A1 and B1 present 2 AFDS aluminum alloy 6061 depositions via photographs. The linear translation speed of sample 1 (Fig. 3A1) is half of the speed of sample 2 (Fig. 3B1). All other input parameters are kept identical during the deposition processes. On the samples,

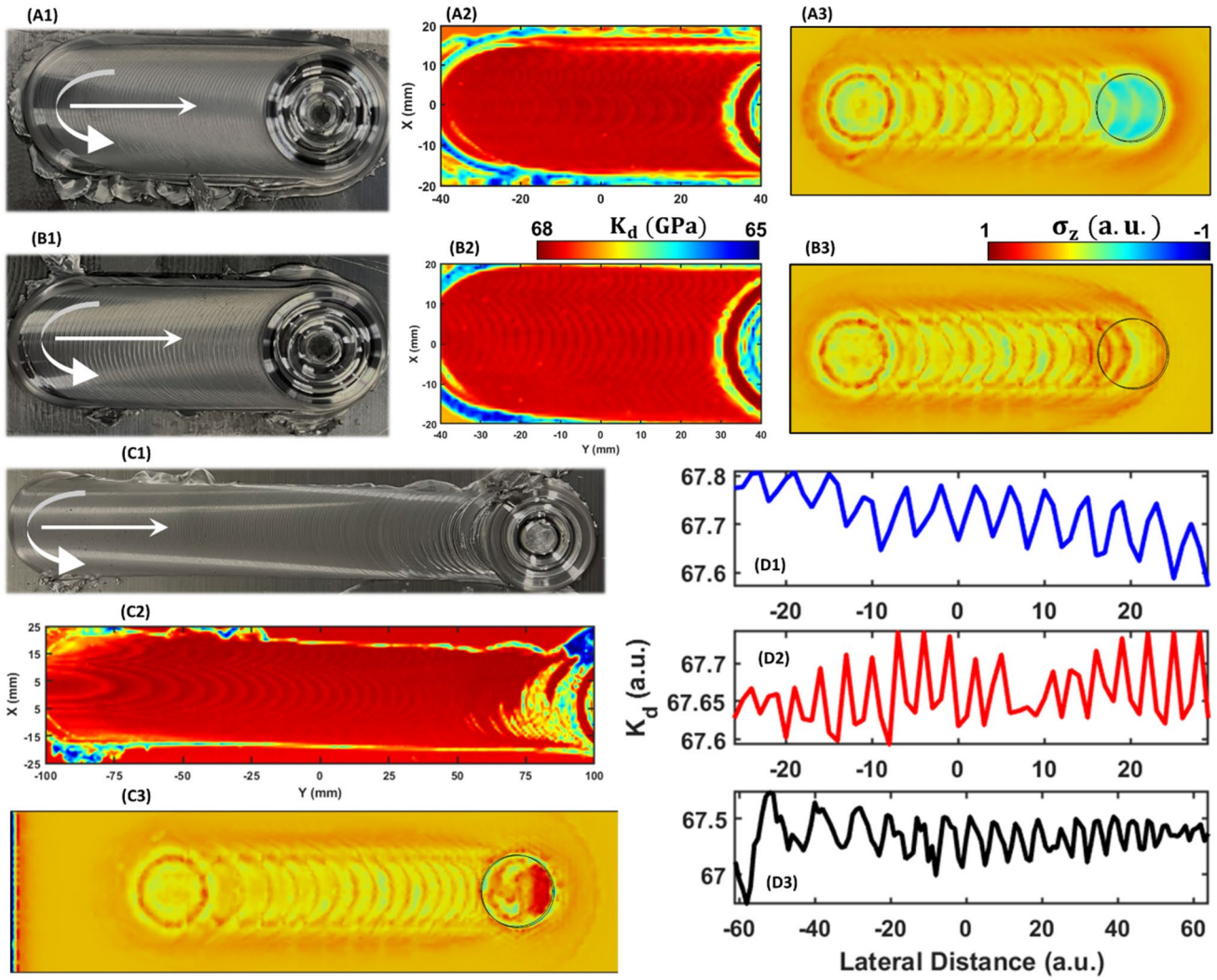

Fig. 3 **A1** and **B1** Photographs of 2 AFDS AA 6061 samples with linear speeds 10 IPM and 12 IPM respectively and identical other parameters. **A2** and **B2** Ultrasound-scanned elastographs of the samples in A1 and B1 showed low ripple radius/ frequency and high ripple radius/ frequency, respectively. **A3** and **B3** Numerical simulation stress distributions with a representative model of AFSD using slow and fast linear motion speeds, respectively. **A3**, **B3**, and **C3** Photograph, experimental elastograph, and simulated stress-field of an AFDS AA 6061 sample with gradient increased linear speeds from 8 to 14 IPM. **D1**, **D2**, and **D3** Central cut-lines from **A2**, **B2**, and **C2** show the well-agreed behaviors with respect to the theoretical estimations (Fig. 2D1, D2, and D3)

the deposition with faster linear speed (Fig. 3B1) showed a significantly smaller periodicity (smaller than 1 mm) of the onion ring structures with respect to the lower linear speed deposition sample (Fig. 3A1). Note that the stress ripples were independent of onion rings and internal banded microstructure.

Observing methodology selection is critical to practically observing stress patterns (ripples). Besides unusual synchrotron or neutron diffraction-based techniques, ultrasound-estimated deviation on the dynamic (effective) bulk modulus is an excellent alternative to observe stress-induced viscoelasticity ripples. Based on the principle of atomic distance-dependent elastic bulk modulus, the unreleased stress or deformation can shift the amplitude of bulk modulus as [5]:

$$K_d \propto a_0 \frac{dF}{dr} \tag{5}$$

where a_0 refers to an original atomic distance, F is the applied force, and r indicates the change in the atomic distance.

In Fig. 3A2 and B2, the ultrasonic scans obtained dynamic bulk modulus elastography results are depicted, showing the distributions of dynamic bulk modulus of samples 1 and 2 (Fig. 3A1 and B1) [19, 47, 48]:

$$K_D = cZ_0\left(\frac{-1-\frac{p_1}{p_e-p_0}-\sqrt{4\frac{p_1}{p_e-p_0}+1}}{\frac{p_1}{p_e-p_0}-2}\right) \tag{6}$$

where p_e is the maximum absolute magnitude of the emitted pulse from the transducer, p_0 and p_1 are the maximum absolute magnitude of reflection from the front and back surface of the sample, t_f and t_i are the start point and endpoint of the pulse, which is determined from an automatic algorithm, c is the speed of sound in the sample, and Z_0 is the reference acoustic impedance of the ambient material.

From the elastographs, it was obvious that the ripple patterns were presented on the elastographs with opposite curvature directions with respect to the onion ring structure on the surface of the samples. In the deposition regions of both samples, the viscoelastic patterns were formed by three regions. The central groove [49, 50] has a width of around 10 mm, presented along the Y-axis in Fig. 3A2. Two minor grooves contain side-wing ripples beside the central major groove located at $X = -10$ and 10 mm along the Y-axis. The periodicity of the side-wing ripples is approximately equal to those in the central regions. Such viscoelasticity-induced residual stress patterns are similar to the scratching deformation patterns in viscoelastic polymers after high strain rate scratching tests in the literature. Noted that, due to the counter-clockwise motion of the tool along the linear motion, the asymmetric stress distribution results in a sharper stress-induced ripple on the retreating side ($X = 10$ mm). In sample 2, the general viscoelastic behaviors were comparable with sample 1. The central groove shows a slightly broader region (~20 mm wide) with a much closer periodicity of the ripples. The minor grooves show the side-wing ripples with a higher repetitive rate concerning sample 1.

In Fig. 3A3 and B3, additional verifications were conducted using a solid mechanics numerical simulation approach. In Fig. 3B3, the setting linear translation (from left to right) speed was doubled with respect to the linear speed in the case in Fig. 3A3. The tool's heating expansion and normal stress produce a Hertzian-liked contact stress pattern on the substrate (left side of Fig. 3A3 and B3). Along with the linear translation of the heated tool, the substrate exhibits viscoelastic and viscoplastic properties that present and hold the ripple patterns along the agreed direction with the theoretical estimation and experimental observations. These patterns are governed by strain hardening and thermal softening behaviors, according to the Johnson-Cook equation [51]:

$$\sigma_{eff} = (A + B(\epsilon)^n)\left(1 + C\ln\left(\frac{\dot{\epsilon}}{\dot{\epsilon}_0}\right)\right)\left(1 - \left(\frac{T - T_0}{T_{melt} - T_0}\right)^m\right) \tag{7}$$

where σ_{eff} refers to effective stress. Factors A, B, C, m, and n are the temperature-dependent yield stress, strain hardening modulus, strain rate sensitivity, thermal softening coefficient, and strain hardening coefficient, respectively. T_{melt} refers to the melting temperature of AA6061 [22, 44]. The Johnson-Cook equation is suitable to present the resulting effective stress distribution affected by the strain induced hardening. Collaborating with the effective viscosity-induced viscoelastic damping effects, the linear "scratching" induced stress ripple can be presented and held on the stress distribution maps (Fig. 3A1, B3, and C3). With a faster linear translation speed in Fig. 3B3, the simulated model shows three more ripples with respect to the slower linear translation case (Fig. 3B3) in the identical processing length, which is agreed with the estimation from periodical G'sell-Jonas equation.

In addition, to present the dependence between linear translation speed and the periodicity of viscoelastic stress ripples, another AFSD deposition sample with a gradient increases linear translation speed (Fig. 3C1). In the elastography (Fig. 3C2), it is evident that the periodicity of the stress ripples decreases from the beginning to the end point of the deposition, where the linear translation speed gradient increases. Compared with the consistent periodicities in the previous cases (Fig. 3A2 and B2), the variation in the periodic stress ripples shows experimental proof of the inverse dependence between ripple periodicity and the linear translation velocity. In Fig. 3C3, a numerical simulation model is conducted with the gradient increasing linear translation speed (1× to 3×) from the left to right side on the substrate. A significant reduction and shortening in the pattern spacing can be observed from the simulation, showing agreement with the experimental elastography. In Fig. 3D1, D2, and D3, a comparison is conducted between the central cutting lines (along the Y-axis) of the elastography of all three samples (Fig. 3A2, B2, and C2). The slower deposition (Fig. 3D1) shows broader and fewer stress ripples with respect to the faster deposition (Fig. 3D2), where gradient speed deposition presents a gradient transition of the shortening and closing of the stress patterns. Such experimentally obtained comparison qualitatively matches the theoretical estimated in the previous section (Fig. 2D1, D2, and D3).

4 Conclusion

This work studied the tool scratching-induced periodical residual stress ripples in the additive friction stir deposition process of aluminum using analytical discussions, numerical simulations, and experimental observations. During the deposition, the softened aluminum has strong viscoelastic damping effects under the normal force and linear motion of the deposition tool, which results in periodic residual stress

ripples on the surface and subsurface regions. The behavior is highly related to the softened aluminum's effective viscosity, strain rate, and temperature, which are dependent on the additive friction stir deposition. The stress ripples have opposite directions and significantly wider periodicity with respect to the onion ring structure. The periodicity of the stress ripples has an inverse proportion with respect to the tool's linear motion speed. The effective radius of the stress ripples is proportional to the tool's linear motion speed.

Author contribution Conceptualization: YJ, TW; methodology: YT, TY, SD; visualization: YJ, TY, and TW; software: YJ, TW, validation: YJ, TY, SD, AN, ND; investigation: YJ, TW, AN, ND; supervision: AN, ND; resources: AN, ND; project administration and fund acquisition: AN, ND.

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Declarations

Conflict of interest The authors have no conflict of interest or competing interests related to this work.

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